



TRS-80 CoCo 3 — GIME & Pinouts

GIME TCC1014 · 6809E · 6821 PIA · MEMORY

DIP pinouts from the top, notch up, pin 1 top-left. Power **red**, ground grey, clocks blue. Verify against the datasheet before powering up.

Legend — reading the pinouts notation

/NAME Active LOW — true / asserted when the pin is at logic **LOW** (~0 V). A leading / (datasheets often use an overline) marks these: e.g. /RES /IRQ /CS /WE.

NAME Active HIGH / normal — asserted when the pin is **HIGH** (~+5 V). No slash. R/W is a level: HIGH = read, LOW = write.

Colour key: **Vcc / Vdd = power** · **GND / Vss = ground** · ϕ = **clock**. **Orientation:** chip seen from the top, notch up — pin 1 top-left, numbers run **down the left** then **up the right**.

GIME TCC1014 68-PLCC pins by function

A0-A3	25, 26, 27, 28
A4-A15	43-54
CPUD0-D7	12-19
RAMD0-D7	64,63,62,61,60,59,58,57
Z0-Z8 mux	32,33,34,36,37,38,40,41,42
/RAS /CAS	4, 5
/WE0 /WE1	10, 11
R/W /RESET	8, 9
E / Q clk	6, 7
EXTAL XTAL	2, 3
/IRQ /FIRQ	21, 20
/ILO /IL1 /IL2	22, 23, 24
S0 S1 S2	31, 30, 29
R G B	68, 67, 66
CVIDEO	65
HSYNC VSYNC	56, 55
VDD / TEST	35 / 39
VSS	1

Advanced Color Video Chip (ACVC): replaces both the 6847 VDG and the 6883 SAM. Hi-res video, 512K MMU, DRAM control, interrupts. Pin map verified from the CoCo3 service-manual schematic (68-pin PLCC — pins follow the package, not a DIP). **Fails:** dead machine, no/bad video.

MC6809E CPU 40-DIP

Vss	1	40	/HALT
/NMI	2	39	TSC
/IRQ	3	38	LIC
/FIRQ	4	37	/RESET
BS	5	36	AVMA
BA	6	35	Q
Vcc	7	34	E
A0	8	33	BUSY
A1	9	32	R/W
A2	10	31	D0
A3	11	30	D1
A4	12	29	D2
A5	13	28	D3
A6	14	27	D4
A7	15	26	D5
A8	16	25	D6
A9	17	24	D7
A10	18	23	A15
A11	19	22	A14
A12	20	21	A13

Same CPU as CoCo 1/2, @ 0.89 or 1.79 MHz. The GIME now supplies its clocks (E, Q) and all timing. **Fails:** dead machine, no boot.

MC6821 PIA x2 40-DIP

Vss	1	40	CA1
PA0	2	39	CA2
PA1	3	38	/IRQA
PA2	4	37	/IRQB
PA3	5	36	RS0
PA4	6	35	RS1
PA5	7	34	/RES
PA6	8	33	D0
PA7	9	32	D1
PB0	10	31	D2
PB1	11	30	D3
PB2	12	29	D4
PB3	13	28	D5
PB4	14	27	D6
PB5	15	26	D7
PB6	16	25	ϕ 2/E
PB7	17	24	CS1
CB1	18	23	/CS2
CB2	19	22	CS0
Vcc	20	21	R/W

Two PIAs as on the CoCo 1/2: U8 keyboard/joystick, U4 sound/cassette/serial. **Fails:** dead keys, no sound/cassette.

Memory & ROM RAM/ROM

RAM	128KB or 512KB (41256/4464 DRAM)
ROM	32KB Super Extended Color BASIC
Video RAM	banked in main RAM by the GIME
MMU	inside the GIME (8 banks × 8KB)

The GIME's MMU pages the 512K into the 6809's 64K space. No separate SAM/VDG on the CoCo3.

Test & Repair tips

Dead / no video	GIME / 6809E / ROM
Bad hi-res only	GIME
Dead keys	U8 PIA
No sound/tape	U4 PIA
Bad RAM	41256/4464 banks

Almost everything routes through the GIME — it's the first suspect for video or dead-machine faults.

Notes verify these

GIME	68-PLCC — pins follow the package, not a DIP
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(GIME, 6809E, 6821 verified)
Amber pins and **flagged chips** are my best reconstruction from incomplete/garbled sources — confirm each against a datasheet before use.